

Ethics and Responsible AI Usage Log: Milestone 3

Ethics decisions and use of AI tools, Modelling, Evaluation and Deployment

BIN381 SDHSA project, Group P1. Log lead: Timothy Wubbeling (602543), with Alexander Solomon and Rudi Jan du Plessis. October 2026

1. Purpose

The project brief asks for ethics to be handled throughout the lifecycle rather than once at the end, and for all use of AI to be acknowledged. This log records both for Milestone 3. Section 2 lists the ethics decisions taken while modelling, evaluating and deploying the model. Section 3 records every use of AI tools: what the tool produced, how the output was checked and what the team decided. Section 4 is the acknowledgement statement for the report, and Section 5 describes how the AI tools were used responsibly.

2. Ethics decisions (Milestone 3)

ID	Date	Stage	Issue	Decision and mitigation	Owner
E3-01	1 Oct	Modelling	Leakage: the eight food-insecurity items define the target	Excluded from the predictors and checked in code (stop if not on the predictor list)	A
E3-02	1 Oct	Modelling	Class imbalance (69.5% secure, 30.5% insecure)	Stratified 5-fold cross-validation; class weights compared with plain stratification	A
E3-03	2 Oct	Evaluation	Accuracy is misleading for an imbalanced, high-stakes screen	Recall, precision, PR-AUC and calibration reported; threshold set on cross-validation predictions for 80% recall, never on the test set	B
E3-04	2 Oct	Evaluation	The best-scoring model was not the most suitable	Binary logistic regression chosen over the random forest for interpretability and calibration, and the trade-off stated openly	B
E3-05	2 Oct	Evaluation	Unweighted results may not represent South African households	Survey-weighted check: at-risk rate 28.8% weighted against 30.6% unweighted; recall 0.783 against 0.807	B
E3-06	2–5 Oct	Interpretation	Grant receipt is associated with higher risk (odds ratio 1.20)	Reported as an association that reflects grant targeting, not a cause; the same wording is used in the app	B, C

ID	Date	Stage	Issue	Decision and mitigation	Owner
E3-07	5 Oct	Fairness	Unequal recall between groups	Equal-opportunity criterion with a 10-point tolerance set before the results were examined; four groups fall outside (incomes above R11,500, no grant recipients, incomes below R2,500, households of seven or more). Disclosed in the app, audit sample added, no group-specific thresholds	C
E3-08	5 Oct	Deployment	Risk of the model being used to decide eligibility	Screening and prioritisation only; an official reviews every higher-risk result (POPIA s71); a caveat on every result and download	C
E3-09	5 Oct	Deployment	Privacy of real households	Uploads processed in memory and not stored; HouseholdID never shown or returned; seven fields only; monitoring log keeps totals only; South African hosting behind sign-in	C
E3-10	5 Oct	Deployment	Stigmatising labels	The app says 'Higher risk: refer for review', never 'food insecure'	C
E3-11	5 Oct	Monitoring	Feedback loop: only flagged households produce new labels	A random audit sample of lower-risk households is followed up every month	C
E3-12	5 Oct	Monitoring	Inflation makes the model under-flag over time	Drift alerts and retraining triggers; shown in a simulated drift test (share flagged fell from 59.8% to 51.9% with a 25% nominal income rise)	C
E3-13	5 Oct	Reproducibility	Packages could not be installed on Windows without Rtools, so others could not reproduce the work	Three package versions in <code>renv.lock</code> moved to their next patch release, which has Windows binaries; the full pipeline was re-run from a clean session	C

Table 1. Ethics decisions in Milestone 3 (A = Alexander Solomon, B = Rudi Jan du Plessis, C = Timothy Wubbeling). E3-01 to E3-06 are taken from the Person A and Person B notebooks.

3. Use of AI tools

Timothy Wubbeling (Person C) used Claude, a generative AI assistant made by Anthropic, through the Claude desktop application. The assistant could read and write files in Timothy's BIN381 project folder and run R code there, so its output was tested on the project's own data rather than produced in isolation. The table records each use.

#	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-1	29 Sep 2026 Timothy (C) Claude (Anthropic)	Planning. Turned the Milestone 3 work split into a step-by-step plan for Person C (an HTML guide with a timeline, tasks and draft text). Flagged three places where the work split contradicted Timothy's Milestone 2 findings (20,940 households, not 20,684; household sizes 8 and 9 valid; zero income valid).	Compared with the work split, the Milestone 1 and 2 reports and the data dictionary.	Timothy used the plan to organise his work and agreed the hand-off with A and B.
AI-2	29 Sep 2026 Timothy (C) Claude (Anthropic)	Starter code for the deployment component: Claude worked along side Person C to create the shiny app and make sure that everything ran correctly. It also helped to make sure that all past work integrated flawlessly, mainly milestones 2 work.	Run against the Milestone 2 cleaned data; automated tests passed; app launched and tested.	Validation rules taken from Timothy's own Milestone 2 cleaning notebook.
AI-3	29 Sep 2026 Timothy (C) Claude (Anthropic)	Repository set-up. Folder structure, .gitignore, README template; moved the existing uploads into data/raw/ and notebook/; explained Git and GitHub Desktop. This helped to share ongoing updates with the team.	Changes reviewed in GitHub Desktop and on GitHub.	Timothy approved each push to the group repository.
AI-4	29 Sep 2026 Timothy (C) Claude (Anthropic)	Explanation. Claude helped me to understand the flows and gave guiders on how to set up the diagrams and flow logic	Not applicable (for understanding only).	Not submitted.

#	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-5	5 Oct 2026 Timothy (C) Claude (Anthropic)	Connecting the final model. When integrating in persons A's work, I ran into some bugs and input miss matches, and Claude helped me to see what I couldn't and resolved them for me.	App probabilities equal Person B's on all 6,279 test households (largest difference 0); recall 0.807 and 58.9% flagged match Person B.	Timothy changed, reviewed and approved the changes before they were pushed.
AI-6	5 Oct 2026 Timothy (C) Claude (Anthropic)	Reproducibility. I ran into some package problems so I asked for Claudes help and it fixed <code>renv.lock</code> for Windows (three packages. Claude worked alongside person C to write my notebook and merge all other Milestone 3 notebooks. Claude also added one line to Person A's notebook, so it knits from notebook/, and added Timothy's Milestone 2 cleaning notebook to the repository.	Notebooks knitted; the Milestone 2 notebook reproduces the cleaned files byte for byte; the merged Milestone 3 notebook ran end to end from a clean session (15 minutes) and reproduced every model and evaluation output (only recorded training times differ; the re-fitted final model gives identical predictions).	Changes to Person A's notebook and to <code>renv.lock</code> were confirmed with Alexander.
AI-7	5 Oct 2026 Timothy (C) Claude (Anthropic)	Report drafting. Help to guide and direct the drafting of Person C report sections (3.5 and 4.1–4.8), placed the app screenshots, helped to formulate the workflow diagram and gave pointer for this log.	Every number traced to data/evaluation/ or the Person C notebook.	Timothy wrote, reviewed and edited the wording before it went to the compiler.

#	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-8	5 Oct 2026 Timothy (C) Claude (Anthropic)	Report compilation. Helped to word Milestone 3 report correctly; a first draft of the Task 1 section written from Person A's notebook and its outputs; Person B's Task 2 and 3 draft brought in unchanged in wording; Person C sections adapted to build on Person B's tables; merged references; and the model change log (models/CHANGELOG.md).	Every Task 1 number taken from Person A's notebook output or the saved models; Person B's text extracted programmatically, not retyped; the PDF checked page by page.	Alexander to review and edit the Task 1 draft; Rudi to review the compiled report before submission.
AI-A1	30 Sep 2026 Alexander (A) Claude (Anthropic)	Work split review. Compared the team's Milestone 3 work split with the brief and with the verified Milestone 2 files. Flagged four errors: a household count of about 20,684 (Milestone 2 established 20,940, of which 20,925 are modelled); a rule turning household sizes 0, 8 and 9 into missing (only 0 is invalid); the income transform described as undecided although Income_Log and Income_PerCapita already exist; and GeoTypeCode described as binary (it has three levels).	Every figure and code meaning checked against M2_final_modeling_dataset.csv and the Milestone 2 data dictionary with a script.	Alexander accepted the four corrections and used the corrected split to plan Task 1.
AI-A2	30 Sep 2026 Alexander (A) Claude (Anthropic)	Explanation. Explained R tidymodels concepts and the R error messages he met.	Not applicable (for understanding only).	Not submitted.
AI-A3	30 Sep 2026 Alexander (A) Claude (Anthropic)	Repository review. Read Person C's files to list what Person C still needed. Found that score.R, fairness_check.R and check_app_matches_model.R expect a single binary output column (.pred_insecure), while the Task 1 models predict three tiers.	Found by reading the code and confirmed against the live repository (the column name is set in app/R/score.R).	The mismatch was resolved with the binary model in AI-A4.

#	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-A4	1 Oct 2026 Alexander (A) Claude (Anthropic)	Binary model for the app. Added a section to the notebook that fits a regularised logistic regression on Food_Insecure_Binary, with the levels relabelled so that the output column is exactly .pred_insecure, and exports it as final_workflow_binary.rds.	stopifnot confirms that .pred_insecure exists before export. Alexander ran the section in RStudio.	Alexander ran and exported the model and pushed it to the repository. The three-tier models remained the Task 1 models; the final model and the threshold were Person B's decisions.
AI-A5	1 Oct 2026 Alexander (A) Claude (Anthropic)	Notebook editing and repository set-up. Shortened and simplified the notebook's explanatory text and comments before it went into the repository (the code was not changed). Explained how to copy files into the cloned repository, commit and push with GitHub Desktop, and how to create renv.lock with renv::init().	Chunk fences counted by script; Alexander re-knitted the notebook. The live repository was re-checked for the data folders, the notebook version and renv.lock (which lists tidymodels, shiny, glmnet, ranger, rpart and knitr).	Alexander made every commit and push himself. The first renv.lock, made in the wrong folder, was not committed; the one in the repository was made in the repository project and was later adjusted for Windows by Timothy (AI-6).

#	Date, person, tool	Task and what the AI produced	How it was checked	Decisions and changes by the team
AI-A6	1 Oct 2026 Alexander (A) Claude (Anthropic)	Report section (Task 1). Drafted the Task 1 section of the report as a Word document: problem framing, data split, predictors, the three techniques and their assumptions, validation design, final models, cross-validation results and reproducibility. Where a number was not yet available the assistant left a marked gap instead of estimating it.	Hyperparameters taken from <code>model_parameters_table.csv</code> ; the cross-validated ROC-AUC values (forest 0.673, logistic 0.669, tree 0.658) and the multicollinearity check (R-squared 0.064, VIF 1.07) pasted from Alexander's own R Studio output; class counts from the Milestone 2 dataset; document rendered and read.	Alexander reviewed the text and the group used it as the starting point for the compiled report. Which model is best was left to Person B's evaluation.
AI-B	2–5 Oct 2026 Rudi (B) Claude (Anthropic)	Evaluation and interpretation. Coding and writing assistant for Tasks 2 and 3: used to plan the Person B tasks, to draft and debug R code in the evaluation notebook, and to help draft the text of the Task 2 and Task 3 sections.	Every number, table and figure in these sections was generated by running the notebook on the team's data; the text was checked against those outputs before submission.	Rudi reviewed and approved all analysis decisions, including the choice of metrics, the threshold rule and the final model. Responsibility for the content rests with the author.

Table 2. Responsible AI usage log for Milestone 3.

4. Acknowledgement statement for the report

The following paragraph is the AI acknowledgement section of the compiled Milestone 3 report .

Acknowledgement of AI use. Generative AI was used in this milestone. Timothy Wubbeling (Person C) used Claude (Anthropic), through the Claude desktop application, as a coding and writing assistant. It was used to plan the Person C tasks; to give guidance on the draft and debug the R code for the Shiny app (input validation, scoring and explanations, drift monitoring), the consistency and fairness checks and the Person C and merged notebooks; to set up the repository and fix the package environment; to guide Timothy on how to create the draft of the deployment, monitoring and ethics sections, and the workflow diagram. All AI-assisted code was run and tested on the project data, and every reported number was checked against the notebooks' outputs. Rudi du Plessis (Tasks 2 and 3) used Claude (Anthropic) as a coding and

writing assistant. It was used to plan the Person B tasks; to draft and debug R code in the evaluation notebook and helped draft the text of the Task 2 and Task 3 sections. The author reviewed and approved all analysis decisions, including the choice of metrics, the threshold rule and the final model. Every number, table and figure in these sections was generated by running the notebook on the team's data. The text was checked against those outputs before submission. Responsibility for the content rests with the author. Alexander Solomon (Task 1) used Claude (Anthropic) as a coding and writing assistant. It was used to check the Milestone 3 work split against the verified Milestone 2 output; to explain tidymodels concepts and R error messages; to review the repository for what the deployment needed; to add the binary screening model to the modelling notebook; to simplify the notebook's explanatory text and set up the repository and renv; and to draft the Task 1 section of the report. Every figure was checked against the Milestone 2 dataset and his own R output, he ran and committed all code himself, and the choice of the final model was left to Person B's evaluation. The group takes full responsibility for the submitted work. Each use is recorded in the group's Ethics and Responsible AI Usage Log.

5. How the AI tools were used responsibly

- Data shared. The assistant worked inside Timothy's project folder, which contains the de-identified Stats SA survey extracts used by the whole class (no names, addresses or contact details) and the group's code and reports. No other personal information was shared.
- Verification, not trust. Code was accepted only after it ran on the project data and passed tests, and numbers were accepted only when they matched the evaluation files. The checks caught errors in both directions: the assistant flagged figures in the work split that Milestone 2 had disproved, and testing caught the assistant's own mistakes (a rounding error that reported recall as 0.809 instead of 0.807, a drift test that first used an unrepresentative sample, and a fairness re-check that compared values at different precision), all corrected before use.
- Human decisions. Choices about the model, the metric and the threshold were made by Persons A and B in their own notebooks. Person C's design choices that the assistant backed up (for example, the two-missing-inputs rule, the 10-point fairness tolerance and the hosting recommendation) were accepted by Timothy, and are open to the group's review.
- Attribution. AI assistance is acknowledged in this log and in the report, not in the repository's commit history; all commits were made under the group members' own accounts after review.
- Academic integrity. The analysis, the decisions and the final wording are the group's responsibility. AI-drafted text proposals were reviewed and edited by the group member responsible for that section before submission.